

Громов Роман, группа ЗМО-РБД-2
Лабораторная работа №3

Список опций утилит tracert

```
C:\Windows\system32\cmd.exe

C:\Users\redwolf>tracert /?

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
              [-R] [-S srcaddr] [-4] [-6] target_name

Options:
  -d          Do not resolve addresses to hostnames.
  -h maximum_hops  Maximum number of hops to search for target.
  -j host-list  Loose source route along host-list (IPv4-only).
  -w timeout    Wait timeout milliseconds for each reply.
  -R          Trace round-trip path (IPv6-only).
  -S srcaddr    Source address to use (IPv6-only).
  -4          Force using IPv4.
  -6          Force using IPv6.

C:\Users\redwolf>
```

Выполнение утилит по заданию:

```
C:\Users\redwolf>hostname
EB4B
```

```
C:\Users\redwolf>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . : localdomain
    IPv6 Address. . . . . : fdb2:2c26:f4e4:0:b92f:183d:3116:23e3
    Temporary IPv6 Address. . . . . : fdb2:2c26:f4e4:0:acde:ce05:2161:998c
    Link-local IPv6 Address . . . . . : fe80::4d49:3d1c:1e7d:21e3%6
    IPv4 Address. . . . . : 10.211.55.19
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.211.55.1

C:\Users\redwolf>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : EB4B
    Primary Dns Suffix . : 
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : localdomain

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . : localdomain
    Description . . . . . : Parallels VirtIO Ethernet Adapter
```

```

C:\Users\redwolf>ping google.com

Pinging google.com [64.233.161.100] with 32 bytes of data:
Reply from 64.233.161.100: bytes=32 time=40ms TTL=128
Reply from 64.233.161.100: bytes=32 time=49ms TTL=128
Reply from 64.233.161.100: bytes=32 time=55ms TTL=128
Reply from 64.233.161.100: bytes=32 time=48ms TTL=128

Ping statistics for 64.233.161.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 40ms, Maximum = 55ms, Average = 48ms

C:\Users\redwolf>ping -n 5 yandex.ru

Pinging yandex.ru [77.88.55.88] with 32 bytes of data:
Reply from 77.88.55.88: bytes=32 time=37ms TTL=128
Reply from 77.88.55.88: bytes=32 time=57ms TTL=128
Reply from 77.88.55.88: bytes=32 time=45ms TTL=128
Reply from 77.88.55.88: bytes=32 time=55ms TTL=128
Reply from 77.88.55.88: bytes=32 time=38ms TTL=128

Ping statistics for 77.88.55.88:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 37ms, Maximum = 57ms, Average = 46ms

```

```

C:\Windows\system32\cmd.exe
C:\Users\redwolf>netstat -an

Active Connections

Proto Local Address           Foreign Address         State
TCP   0.0.0.0:135              0.0.0.0:0              LISTENING
TCP   0.0.0.0:445              0.0.0.0:0              LISTENING
TCP   0.0.0.0:5040             0.0.0.0:0              LISTENING
TCP   0.0.0.0:49664            0.0.0.0:0              LISTENING
TCP   0.0.0.0:49665            0.0.0.0:0              LISTENING
TCP   0.0.0.0:49666            0.0.0.0:0              LISTENING
TCP   0.0.0.0:49667            0.0.0.0:0              LISTENING
TCP   0.0.0.0:49669            0.0.0.0:0              LISTENING
TCP   0.0.0.0:49676            0.0.0.0:0              LISTENING
TCP   10.211.55.19:139         0.0.0.0:0              LISTENING
TCP   10.211.55.19:49411       4.207.247.137:443      ESTABLISHED
TCP   10.211.55.19:52972       4.207.247.137:443      ESTABLISHED
TCP   10.211.55.19:53013       62.115.252.11:443      CLOSE_WAIT
TCP   10.211.55.19:53015       62.115.252.11:443      CLOSE_WAIT
TCP   10.211.55.19:53019       62.115.252.11:443      CLOSE_WAIT
TCP   10.211.55.19:53105       199.232.210.172:80     TIME_WAIT
TCP   10.211.55.19:53110       57.155.101.212:443     TIME_WAIT
TCP   10.211.55.19:53111       20.190.147.0:443       TIME_WAIT
TCP   10.211.55.19:53112       48.209.109.33:443      ESTABLISHED
TCP   10.211.55.19:53113       85.210.193.152:443     TIME_WAIT
TCP   10.211.55.19:53115       23.11.40.157:80        ESTABLISHED
TCP   10.211.55.19:53117       20.50.73.13:443        TIME_WAIT
TCP   10.211.55.19:53118       57.155.101.212:443     TIME_WAIT
TCP   10.211.55.19:53120       23.11.40.157:80        ESTABLISHED
TCP   10.211.55.19:53121       199.232.210.172:80     TIME_WAIT

```

```

C:\Users\redwolf>tracert google.com

Tracing route to google.com [64.233.161.100]
over a maximum of 30 hops:

  1    38 ms    38 ms    38 ms  lh-in-f100.1e100.net [64.233.161.100]

Trace complete.

```

```

C:\Users\redwolf>pathping google.com

Tracing route to google.com [64.233.161.100]
over a maximum of 30 hops:
  0  EB4B.localdomain [10.211.55.19]
  1  lh-in-f100.1e100.net [64.233.161.100]

Computing statistics for 25 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
   0                               0/ 100 = 0%      EB4B.localdomain [10.211.55.19]
   1   54ms    0/ 100 = 0%      0/ 100 = 0%      lh-in-f100.1e100.net [64.233.161.100]

Trace complete.

```

```

C:\Users\redwolf>arp -a

Interface: 10.211.55.19 --- 0x6

   Internet Address      Physical Address      Type
10.211.55.1              00-1c-42-00-00-18     dynamic
10.211.55.255            ff-ff-ff-ff-ff-ff     static
224.0.0.22               01-00-5e-00-00-16     static
224.0.0.251              01-00-5e-00-00-fb     static
224.0.0.252              01-00-5e-00-00-fc     static
239.255.255.250          01-00-5e-7f-ff-fa     static
255.255.255.255          ff-ff-ff-ff-ff-ff     static

```

```

C:\Users\redwolf>net view
System error 6118 has occurred.

The list of servers for this workgroup is not currently available

```

```

C:\Users\redwolf>nslookup yandex.ru
Server:  pr1-local-ns-server.shared
Address:  10.211.55.1

Non-authoritative answer:
Name:     yandex.ru
Addresses: 77.88.44.55
           77.88.55.88
           5.255.255.77

```

Анализ ipconfig /all

```
C:\Users\redwolf>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : EB4B
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : localdomain

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . : localdomain
    Description . . . . . : Parallels VirtIO Ethernet Adapter
    Physical Address. . . . . : 00-1C-42-45-EA-71
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv6 Address. . . . . : fdb2:2c26:f4e4:0:b92f:183d:3116:23e3(Preferred)
    Temporary IPv6 Address. . . . . : fdb2:2c26:f4e4:0:acde:ce05:2161:998c(Preferred)
    Link-local IPv6 Address . . . . . : fe80::4d49:3d1c:1e7d:21e3%6(Preferred)
    IPv4 Address. . . . . : 10.211.55.19(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Wednesday, May 20, 2026 2:19:10 PM
    Lease Expires . . . . . : Wednesday, May 20, 2026 2:49:10 PM
    Default Gateway . . . . . : 10.211.55.1
    DHCP Server . . . . . : 10.211.55.1
    DHCPv6 IAID . . . . . : 83893314
    DHCPv6 Client DUID. . . . . : 00-01-00-01-31-0D-E7-79-00-1C-42-45-EA-71
    DNS Servers . . . . . : 10.211.55.1
    NetBIOS over Tcpi. . . . . : Enabled
```

Команда показывает полную конфигурацию TCP/IP для всех сетевых адаптеров, например MAC-адрес , IP-адрес компьютера, маска подсети, основной шлюз (роутер), адрес DHCP-сервера, адреса DNS-серверов.

Альтернативный способ - в Windows через графический интерфейс: Параметры -> Сеть и Интернет -> Свойства сети.

Анализ pathping и отличие от ping и tracert

```
C:\Users\redwolf>ping yandex.ru

Pinging yandex.ru [77.88.55.88] with 32 bytes of data:
Reply from 77.88.55.88: bytes=32 time=38ms TTL=128
Reply from 77.88.55.88: bytes=32 time=45ms TTL=128
Reply from 77.88.55.88: bytes=32 time=40ms TTL=128
Reply from 77.88.55.88: bytes=32 time=38ms TTL=128

Ping statistics for 77.88.55.88:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 38ms, Maximum = 45ms, Average = 40ms

C:\Users\redwolf>tracert yandex.ru

Tracing route to yandex.ru [77.88.55.88]
over a maximum of 30 hops:

  0  189 ms    38 ms    36 ms  yandex.ru [77.88.55.88]

Trace complete.

C:\Users\redwolf>pathping yandex.ru

Tracing route to yandex.ru [77.88.55.88]
over a maximum of 30 hops:
  0  EB4B.localdomain [10.211.55.19]
  1  yandex.ru [77.88.55.88]

Computing statistics for 25 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
     RTT      Lost/Sent = Pct  Lost/Sent = Pct  Lost/Sent = Pct
  0  0ms       0/ 100 = 0%      0/ 100 = 0%      0/ 100 = 0%      EB4B.localdomain [10.211.55.19]
  1  59ms      0/ 100 = 0%      0/ 100 = 0%      0/ 100 = 0%      yandex.ru [77.88.55.88]

Trace complete.
```

В отличие от ping и tracert, pathping не просто показывает маршрут или проверяет доступность узла, а вычисляет процент утерянных пакетов на каждом хопе, выявляя самые медленные и ненадёжные участки маршрута.

Какие утилиты на ваш взгляд являются самыми полезными и почему?

На мой взгляд, одни из самых полезных – это `ipconfig` (чтобы узнать параметры сети), `ping` (для проверки интернета и работы отдельных серверов) и `tracert` (для поиска конкретной проблемной точки).